GMM, MLE and Tests for Non-linear Restrictions

Lecture 7

This lecture introduces Generalized Method of Moments (GMM), Maximum Likelihood (ML) estimation in a regression setting and discusses tests for non-linear restrictions. As it will be clear later, the Maximum Likelihood method is convenient for dealing with nonlinear models and in some cases non-linear hypothesis. For instance, if you have binary outcome (e.g., $y_i = 1$ if the governor was elected and $y_i = 0$ otherwise) and observables variables (e.g., taxes, state's unemployment rate, etc), it can be shown that the conditional mean model is,

$$\mathbb{E}(y|x) = P(y = 1|x) = F(x'\beta).$$

In the next sections, we discuss the method and associated testing procedures.

1 Generalized Method of Moments (GMM)

Before we present the GMM approach, we briefly review a few simple examples of the Method of Moments. Consider $\{y_i\}$ drawn from an *iid* distribution whose first moment exists (e.g., y is not distributed as Cauchy). In the population,

$$\mathbb{E}(y - \mu) = 0$$

so we can use the corresponding sample moment condition

$$\frac{1}{n}\sum_{i=1}^{n}(y_i - \mu) = 0 \Rightarrow \hat{\mu} = \bar{y}$$

Another example that we already discussed is the IV estimator, that can be defined in terms of the moment condition,

$$\mathbb{E}(z_i u_i) = \mathbb{E}(\mathbb{E}(z_i u_i|z)) = \mathbb{E}(z\mathbb{E}(u_i|z)) = \mathbb{E}(z \times 0) = 0,$$

or alternatively,

$$\mathbb{E}(z_i(y_i - x_i'\beta)) = 0,$$

The IV estimator could be obtained as the solution of the corresponding sample moment condition,

$$\frac{1}{n}\sum_{i=1}^{n}(z_i(y_i - x_i'\beta)) = 0,$$

yielding the instrumental variable estimator presented in previous lecture,

$$\hat{\beta} = \left(\sum_{i=1}^{n} z_i x_i'\right)^{-1} \sum_{i=1}^{n} z_i y_i = (Z'X)^{-1}Z'y$$

We generalize the procedure directly considering models that satisfy the condition,

$$\mathbb{E}(g(w_i, \theta_0)) = 0$$

where $g(\cdot)$ is a function in $\mathbb{R}^l$ and the parameter $\theta_0 \in \Theta \subset \mathbb{R}^p$. For example, $g(w_i, \theta_0)$ had the following linear form

$$z_i(y_i - x_i'\beta),$$

with the minimal requirement of p moment conditions (e.g., $l = p$).

Now suppose than we have l moment conditions with $l > p$. The sample moment counterpart

$$g_n = \frac{1}{n}\sum_{i=1}^{n} g(w_i, \theta_0)$$

has no solution, but we can select p linear combinations of the sample moments in order to obtain the estimator. Therefore, given $g_n(\theta)$, one can choose a positive semidefinite matrix W_n and define the estimator that minimizes

$$\hat{\theta}_{GMM} = \arg\min_{\theta\in\Theta} \|g_n(\theta)\|^2_{W_n} = \arg\min_{\theta\in\Theta} g_n(\theta)'W_n g_n(\theta)$$

The matrix $W_n \to W$ as the sample size goes to infinity with W being positive definite. Of course, the matrix W_n is important for estimation.

1.0.1 Example: Ordinary Least Squares

Consider the case where the errors and the independent variables are independent. For the OLS estimator the conditions are,

$$\mathbb{E}(x_i(y_i - x_i'\beta)) = 0$$

The sample moments conditions are,

$$g_n(\beta) = \frac{1}{n}\sum_{i=1}^{n} x_i(y_i - x_i'\beta)$$

The number of moment conditions is equal to the number of unknown parameters. In this case, we say that the model is just or exactly identified. The IV case, illustrated above, also represents a case where a model is exactly identified.

1.0.2 Example: Two Stage Least Squares

For the 2SLS, we may have more moment conditions than parameters (e.g., $\dim(z_i) > \dim(x_i)$)), but a proper matrix W will be enough to obtain the estimator,

$$\arg\min g_n(\beta)' W_n g_n(\beta)$$

with $W_n = (Z'Z)^{-1}$ and $g_n = Z'(y - X\beta)$. We note that

$$X'ZWZ(y - X\beta) = 0,$$

implies

$$\hat{\beta} = (X'ZWZ'X)^{-1}X'ZWZ'Xy = (X'P_zX)^{-1}X'P_zy$$

The solution depends on the matrix W, and of course, the solution of the minimization problem is the same for W's that differ by a constant of proportionality. Note that this is related with the assumption of homocedastic errors. What should be the weighting matrix under heterocedasticity? Let S be the covariance matrix of the moment condition g,

$$S = \frac{1}{n}\mathbb{E}(Z'uu'Z) = \frac{1}{n}Z'\Omega Z$$

To obtain the optimal GMM estimator in this case, we need:

1. Estimate $\hat{\beta}$ by IV methods to obtain $\hat{u}$.
2. Use $\hat{u}$ to find,
$$\hat{W} = \hat{S}^{-1} = \frac{1}{n}(Z'\hat{\Omega}Z)^{-1}$$
where $\hat{\Omega} = \mathrm{diag}(\hat{u}_i^2)$.
3. Find the feasible, optimal GMM as
$$\hat{\beta} = (X'Z(Z'\hat{\Omega}Z)^{-1}Z'X)^{-1}X'Z(Z'\hat{\Omega}Z)^{-1}Z'Xy$$

2 Further Considerations

If the model is identified, the estimator is consistent and asymptotically normally distributed under regularity conditions,

1. The first regularity condition assumes that

$$\frac{\partial g_n(\theta)}{\partial \theta'} \rightarrow \frac{\partial \mathbb{E}(g(\theta))}{\partial \theta'} = G$$

2. Also we assume the standard square-root consistency,

$$\sqrt{n} g_n(\theta) \rightsquigarrow \mathcal{N}(0, S^{-1})$$

Then, the estimator is asymptotically normally distributed,

$$\sqrt{n}(\hat{\theta} - \theta) \rightsquigarrow \mathcal{N}(0, (G'WG)^{-1} G'WSW'G(G'WG)^{-1})$$

If we choose W_n such that $W_n \rightarrow S^{-1} = W$, we have that the asymptotic variance is simply,

$$(G'S^{-1}G)^{-1},$$

the case that it is known as efficient GMM (Hansen 1982) when S is known. This is perhaps a disadvantage of the estimator since requires to know the asymptotic covariance matrix. The solution is to estimate it, but this idea may fail in small and intermediate samples. If the sample size is not large enough, or if the moment conditions is large, the estimation of the asymptotic covariance matrix in the first step can result in bad estimates in the second step. The problem has been documented in the literature, and various solutions (e.g. simultaneous estimation) has been proposed.

3 ML Estimation

Let $\{(y_i, x_i) : i = 1...n\}$ be an independently and identically distributed (*i.i.d.*) sample of the response y and the independent variables x.

We can define the likelihood function as

$$\mathcal{L}(\theta) = f(y, x|\theta),$$

but since our goal is to model the behavior of y in terms of x, we consider directly the conditional density of y_i,

$$f(y_i|x_i, \theta_0)$$

Then we can define the conditional log-likelihood function as

$$\ell(\theta; y_i|x_i) \equiv \log f(y_i|x_i, \theta)$$

Considering an iid sample of n observations $\{(y_i, x_i)\}$, the joint conditional density function is basically

$$\prod_{i=1}^{n} f(y_i|x_i, \theta) = f(y_1|x_1, \theta) \ldots f(y_n|x_n, \theta)$$

and consequently the log-likelihood function is

$$\ell(\theta; y|x) = \log(\prod_{i=1}^{n} f(y_i|x_i, \theta)) \equiv \sum_{i=1}^{n} \log f(y_i|x_i, \theta)$$

Example 1. (Normal Location Scale) Suppose that u_i is distributed as $\mathcal{N}(\beta_0, \sigma^2)$. Then the probability density function is

$$\frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(u_i-\beta_0)^2}{2\sigma^2}}$$

If the observations are independent, we have

$$(2\pi\sigma^2)^{-n/2} e^{-\frac{1}{2\sigma^2}\sum_i (u_i-\beta_0)^2},$$

and consequently the log-likelihood as,

$$\ell(\beta_0, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(u_i - \beta_0)^2$$

Example 2. (The regression model) Consider the classical regression model,

$$y_i = x_i'\beta + u_i$$

where y and x are iid, jointly distributed random variables. Transforming variables will get the log-likelihood as

$$\sum_{i=1}^{n} \log f(y_i|x_i, \theta) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - x_i'\beta)^2$$

where $\theta = (\beta, \sigma^2)$.

Example 3. (Poisson density) Consider now that the dependent variable y_i takes non-negative discrete value (e.g. number of visits to the doctor). The Poisson density function may be a natural direction,

$$f(y|\lambda) = \frac{e^{-\lambda}\lambda^y}{y!}; \qquad y = 0, 1, 2, ...$$

The usual specification in regression analysis is $\lambda = \exp(x'\beta)$, leading to a log-likelihood function of the form

$$\sum_{i=1}^{n} \log f(y_i|x_i, \theta) = -\sum_{i=1}^{n} \exp(x_i'\beta) + \sum_{i=1}^{n} y_i x_i'\beta - \sum_{i=1}^{n} \log y_i!$$

3.1 Maximum Likelihood Estimator

Consider the following assumptions:

1. The model $y_i = m(x_i, u_i, \theta)$ is correctly specified where $m(\cdot)$ is a known function that can be linear (e.g., $y = x'\beta + u$).
2. $\{(y_i, x_i) : i = 1...n\}$ is an i.i.d. sample
3. Θ is a compact subspace in $\mathbb{R}^p$ that contains θ_0
4. The density function $f(y_1|x, \theta)$ is continuous and has continuous second order derivatives over a compact set Θ.
5. The likelihood function has a unique solution.

The MLE principle is to choose an estimator of a parameter of the conditional distribution of y, θ_0, that maximizes the probability of observing the actual sample. The maximum likelihood estimator (MLE) is the estimator that maximizes the (log)likelihood function,

$$\hat{\theta} = \arg\max_{\theta\in\Theta} \ell(\theta; y|x)$$

where Θ is a compact subspace in $\mathbb{R}^p$ that contains θ.

Example 4. (Linear Regression) Consider the log-likelihood function we obtained before using matrix notation,

$$const - \frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}(y - X\beta)'(y - X\beta)$$

Taking the first order condition,

$$\begin{aligned}
\frac{\partial \ell}{\partial \beta} &= \frac{1}{\sigma^2} X'(y - X\beta) = 0 \\
\frac{\partial \ell}{\partial \sigma^2} &= -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}(y - X\beta)'(y - X\beta) = 0
\end{aligned}$$

Solving the system of equations, we find that the MLE estimators are

$$\hat{\beta} = (X'X)^{-1}X'y \quad \text{and} \quad \hat{\sigma}^2 = \frac{1}{n}(y - X\hat{\beta})'(y - X\hat{\beta})$$

We know that the estimator $\hat{\beta}$ is unbiased because it is identical to the OLS estimator. In contrast, the estimator of the variance parameter $\hat{\sigma}^2$ is biased. (Why?). Also,

$$\begin{aligned}
\frac{\partial^2 \ell}{\partial \beta \beta'} &= -\frac{X'X}{\sigma^2} \\
\frac{\partial^2 \ell}{\partial \beta \sigma^2} &= -\frac{X'(y - X\beta)}{\sigma^4} \\
\frac{\partial^2 \ell}{\partial \sigma^2 \beta'} &= -\frac{(y - X\beta)'X}{\sigma^4} \\
\frac{\partial^2 \ell}{\partial \sigma^2 \sigma^2} &= \frac{n}{2\sigma^4} - \frac{(y - X\beta)'(y - X\beta)}{\sigma^6}
\end{aligned}$$

The expectation of the $p \times p$ matrix of second derivatives is called the information matrix,

$$\begin{aligned}
I(\theta_0|X) &= -\mathbb{E}(\partial^2 \ell / \partial\theta\partial\theta' | X) \\
&= -\begin{bmatrix} \mathbb{E}(\partial^2 \ell/\partial\beta\partial\beta') & \mathbb{E}(\partial^2 \ell/\partial\beta\partial\sigma^2) \\ \mathbb{E}(\partial^2 \ell/\partial\sigma^2\partial\beta') & \mathbb{E}(\partial^2 \ell/\partial(\sigma^2)^2) \end{bmatrix} \\
&= \begin{bmatrix} \frac{X'X}{\sigma^2} & 0 \\ 0 & \frac{n}{2\sigma^4} \end{bmatrix}
\end{aligned}$$

Note that the off diagonal elements of the information matrix $I(\theta)$ are zero suggesting that $\hat{\beta}$ and $\hat{\sigma}^2$ are "independent". The matrix of second derivatives is n.s.d.

Example 5. (Poisson density) The Poisson case has no explicit solution, thus numerical methods are useful in this case. By differentiating the likelihood function, we obtain $\hat{\beta}$ as the solution of

$$\sum_{i=1}^{n} y_i x_i - \sum_{i=1}^{n} \exp(x_i \hat{\beta}) x_i = 0$$

The previous case it is common in empirical analysis. Sometimes obtaining the MLE estimator gets complicated not yielding analytic solutions. Below I will briefly discuss alternatives to overcome the problem.

3.2 Efficiency

Let define the score function as,

$$S_i = \frac{\partial \ell_i(\theta)}{\partial \theta} = \left(\left(\frac{\partial \ell_i(\theta)}{\partial \theta_1} \right), ..., \left(\frac{\partial \ell_i(\theta)}{\partial \theta_p} \right) \right)',$$

with $\mathbb{E}(S_i(\theta_0)) = 0$, and the Hessian matrix,

$$H_i = \frac{\partial^2 \ell_i(\theta)}{\partial \theta \partial \theta'} = \frac{\partial S_i}{\partial \theta},$$

a symmetric $p \times p$ matrix of second derivatives. Evaluated at the true parameter θ_0, we have that

$$-\mathbb{E}[H(\theta)] = -\mathbb{E}\left[\frac{\partial^2 \ell(\theta)}{\partial \theta \partial \theta'}\right] = \mathbb{E}\left[\left(\frac{\partial \ell(\theta)}{\partial \theta}\right)\left(\frac{\partial \ell(\theta)}{\partial \theta'}\right)'\right] = I(\theta_0),$$

This property is important for empirical analysis. It is time consuming to compute the second derivative of a likelihood function, but the previous expression suggest an alternative, perhaps easier way: compute the product of the two score functions. We need to be cautious, though. The previous analysis holds when we know the conditional distribution of y.

In the case of iid samples,

$$-\sum_{i=1}^{n} \mathbb{E}\left(\frac{\partial^2 \ell_i(\theta)}{\partial \theta \partial \theta'}\right) = -n\mathbb{E}\left(\frac{\partial^2 \ell_i(\theta)}{\partial \theta \partial \theta'}\right) = n\mathbb{E}\left[\left(\frac{\partial \ell_i(\theta)}{\partial \theta}\right)\left(\frac{\partial \ell_i(\theta)}{\partial \theta'}\right)'\right] = n\Omega(\theta),$$

indicating that the information matrix is n times the information contained in one observation. Note that Ω is the variance of the score function.

It can be shown that the MLE estimator is consistent meaning that

$$\hat{\theta} \to \theta,$$

and asymptotically normal

$$\sqrt{n}(\hat{\theta} - \theta_0) \rightsquigarrow \mathcal{N}(0, \Omega(\theta)^{-1})$$

If the number of observations is sufficiently large, the estimator has asymptotic variance that can be approximated by the inverse of the information contained in one observation. This results shows the efficiency of the MLE estimator, which reaches the Cramer-Rao lower bound (the lower bound of the variance of unbiased estimators),

$$V(\tilde{\theta}) \geq (nI(\theta))^{-1},$$

for any unbiased $\tilde{\theta}$ estimator of θ_0, provided that the information matrix is non-singular. Note that MLE is attractive for estimation of parametric models since it achieves the lower bound.

4 Testing procedures

This section considers tests of linear and non-linear hypothesis. We review three tests based on the likelihood principle and we provide the basis to consider tests when classical methods fails. Suppose we have an iid sample of n observations $\{(y_i, x_i);\ i = 1, ..., n\}$. We consider test of hypothesis of the form,

$$h(\theta) = 0$$

where $h(\cdot)$ is a $r \times 1$ vector function of θ with $r \leq p$. We assume that $H(\theta) = \partial h(\theta)/\partial\theta$, a $p \times r$ matrix, is well defined and it has full column rank

$$\text{rank}(H(\theta)) = r,$$

which means that the restrictions imposed are independent. Before considering the test, let me define a couple of important concepts. We denote the restricted parameter space

$$\Theta_0 = \{\theta \in \Theta | h(\theta) = 0\},$$

implying that that $\Theta_0 \subset \Theta$. Recall that the MLE estimator is defined as,

$$\hat{\theta} = \arg\max_{\theta\in\Theta}\{\ell(\theta)\}$$

Also, we can define the restricted MLE estimator as the argument that maximizes the restricted log-likelihood function,

$$\tilde{\theta} = \arg\max_{\theta\in\Theta_0} \ell(\theta)$$

or equivalently,

$$\max_{\theta\in\Theta_0} \ell(\theta) \quad \text{subject to} \quad h(\theta) = 0$$

4.1 Likelihood Ratio Test

The test is defined as,

$$\text{LR} = 2\{\ell(\hat{\theta}) - \ell(\tilde{\theta})\}$$

The intuition behind the test is that if the null hypothesis is true, the likelihood functions should be similar. Therefore, higher values of LR implies higher chances of being rejecting the null hypothesis H_0.

4.2 Lagrange Multiplier Test

$$\mathrm{LM} = \frac{\partial \ell(\tilde{\theta})}{\partial \theta'} I(\tilde{\theta})^{-1} \frac{\partial \ell(\tilde{\theta})}{\partial \theta},$$

which is motivated by the fact that the score function $\partial \ell(\hat{\theta}/\partial\theta) = 0$. Therefore, if the null hypothesis H_0 is true, we expect that the maximum should occur at $\partial \ell(\tilde{\theta}/\partial\theta) \approx 0$ failing to reject the null.

4.3 Wald Test

The test is defined as,

$$\mathrm{W} = h(\hat{\theta})' \left[H(\hat{\theta})' I(\hat{\theta})^{-1} H(\hat{\theta}) \right]^{-1} h(\hat{\theta})$$

If H_0 is true, $h(\hat{\theta}$ is close to zero. The test statistic is asymptotically χ^2_r where r denotes the rank of the matrix H (i.e. $r = \mathrm{rank}(H)$ -the number of restrictions imposed by H_0-).

Example 6. (Tests of Nonlinear Restrictions) Consider testing

$$\mathrm{H}_0 : h(\theta) = \frac{\theta_1}{\theta_2} - c = 0$$

Then the Wald statistics is

$$\mathrm{W} = \left(\frac{\theta_1}{\theta_2} - c \right) \left[\left(\begin{array}{ccc} \frac{1}{\hat{\theta}_2} & - \frac{\hat{\theta}_1}{\hat{\theta}_2^2} & 0 \end{array} \right) \left[\begin{array}{cccc} \hat{v}_{11} & \hat{v}_{12} & \ldots & \ldots \\ \cdot & \hat{v}_{22} & \ldots & \ldots \\ \ldots & \ldots & \ldots & \end{array} \right] \left(\begin{array}{c} \frac{1}{\hat{\theta}_2} \\ -\frac{\hat{\theta}_1}{\hat{\theta}_2^2} \\ 0 \end{array} \right) \right]^{-1} \left(\frac{\theta_1}{\theta_2} - c \right)$$

where 0 is a $(r-2) \times r$ matrix of zeros and v_{jj} denote the jj-th element of the estimated asymptotic covariance matrix. Under the null W is asymptotically distributed as χ^2_1.

The problem with this test is that it depends on the estimated asymptotic covariance matrix. There are alternatives for non-linear restrictions to be discussed in the next courses (e.g., Delta Method, Bootstrap).

Example 7. (Restricted MLE) Consider now a long-likelihood of the form,

$$\ell(\beta, \sigma^2) = -\frac{n}{2} log(2\pi) - \frac{n}{2} log\sigma^2 - \frac{1}{2\sigma^2}(y - X\beta)'(y - X\beta) + 2\lambda'(R\beta - \delta),$$

where λ is a potentially $p \times 1$ vector. Assuming we know the value of σ^2, the solution can be written as,

$$\frac{\partial \ell}{\partial \beta} = 0 \Rightarrow -2X'y + 2X'X\widetilde{\beta} + 2R'\lambda = 0$$

$$\frac{\partial \ell}{\partial \lambda} = 0 \Rightarrow R\widetilde{\beta} = \delta$$

In other words,

$$\left[\begin{array}{cc} X'X & R' \\ R & 0 \end{array}\right]\left[\begin{array}{c} \widetilde{\beta} \\ \lambda \end{array}\right] = \left[\begin{array}{c} X'y \\ \delta \end{array}\right]$$

Therefore,

$$(\widetilde{\beta}', \lambda)' = \left[\begin{array}{cc} X'X & R' \\ R & 0 \end{array}\right]^{-1}\left(\begin{array}{c} X'y \\ \delta \end{array}\right)$$

Applying inverse-partitioned matrix formulae, we obtain,

$$\begin{aligned}
\widetilde{\beta} &= (X'X)^{-1}X'y - (X'X)^{-1}R'[R(X'X)^{-1}R']^{-1} \\
&\quad R(X'X)^{-1}X'y + (X'X)^{-1}R'(R(X'X)^{-1}R')^{-1}\delta \\
&= \hat{\beta} - (X'X)^{-1}R'(R(X'X)^{-1}R')^{-1}R\hat{\beta} + (X'X)^{-1}R'(R(X'X)^{-1}R')^{-1}\delta \\
&= \hat{\beta} - D(R\hat{\beta} - \delta)
\end{aligned}$$

where $D = (X'X)^{-1}R'(R(X'X)^{-1}R')^{-1}$. Note that

$$E(\widetilde{\beta}) = E(\hat{\beta} - D(R\hat{\beta} - \delta)) = \beta - D(R\beta - \delta) = \beta,$$

if $R\beta - \delta = 0$ (the null hyphotesis is true). If it is false, $\tilde{\beta}$ is biased.

In terms of the variance of $\widetilde{\beta}$, we have that,

$$\begin{aligned}
V(\widetilde{\beta}) &= V(\hat{\beta} - D(R\hat{\beta} - \delta)) = V(\hat{\beta} - DR\hat{\beta}) = V((I - DR)\hat{\beta}) \\
&= (I - DR)V(\hat{\beta})(I - DR)' \\
&= \sigma^2(I - DR)(X'X)^{-1}(I - DR)' \\
&= \sigma^2(X'X)^{-1} - \sigma^2[DR(X'X)^{-1} + (X'X)^{-1}R'D' - DR(X'X)^{-1}R'D']
\end{aligned}$$

Note that,

$$\begin{aligned}
DR(X'X)^{-1} &= (X'X)^{-1}R'(R(X'X)^{-1}R')^{-1}R(X'X)^{-1} \\
&= (X'X)^{-1}R'D'.
\end{aligned}$$

and

$$\begin{aligned} DR(X'X)^{-1}R'D' &= (X'X)^{-1}R'(R(X'X)^{-1}R')^{-1}(R(X'X)^{-1}R')D' \\ &= (X'X)^{-1}R'D' \end{aligned}$$

Therefore,

$$\begin{aligned} V(\widetilde{\beta}) &= \sigma^2(X'X)^{-1} - \sigma^2 DR(X'X)^{-1} \\ &= \sigma^2(X'X)^{-1}(I - DR) \leq V(\hat{\beta}). \end{aligned}$$

The solution for the lagrange multiplier is,

$$\begin{aligned} \widetilde{\lambda} &= (R(X'X)^{-1}R')^{-1}R(X'X)^{-1}X'y - (R(X'X)^{-1}R')^{-1}\delta \\ &= W(R\hat{\beta} - \delta). \end{aligned}$$

It is easy to see that $E(\widetilde{\lambda}) = 0$ under the null and $V(\widetilde{\lambda}) = \sigma^2 W$.

5 Model Selection

In some situations, typically time series, we may not know the dimension of the model (e.g., how many parameters our linear model should have). This important aspect of modeling in statistics and econometrics was early addressed by Akaike (1969) and Schwartz (1978).

In a regression setting, the likelihood can be written as

$$\ell(\beta, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}(y - X\beta)'(y - X\beta)$$

By plugging in $\hat{\beta}$ and $\hat{\sigma}^2 = \mathcal{S}/n$, we obtain the concentrated likelihood function

$$\ell(\hat{\sigma}^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2} - \frac{n}{2}\log\hat{\sigma}^2$$

suggesting that for a given sample size, the likelihood can be increased by adding covariates to the model to reduce the residual sum of squares $\mathcal{S}$.

Akaike (1969) proposed to choose a model from a set that performs well on the basis of forecasting. His proposal is based on the following function,

$$\text{AIC}(j) = \ell_j(\hat{\theta}) - p_j$$

where $\ell_j(\hat{\theta})$ is the likelihood corresponding to the j^{th} model maximized over $\theta \subset \Theta_j$ (typically we have $p_1 < p_2 < ... < p_n$). AIC stands for Akaike Information Criteria. The basic idea of the additional term in AIC, called a penalty term, is to bring a rule over the fit of various specifications by "penalizing" an increase in the number of regressions. Akaike's model selection rule was to evaluate AIC(j) over j models, choosing the model j^* that maximize AIC(j).

A latter criterion, which is known as Schwarz information criterion, is based on a Bayesian approach to the problem analyzed in Akaike (1969). Schwarz proposed

$$\text{SIC}(j) = \ell_j(\hat{\theta}) - \frac{1}{2} p_j \log n$$

Note that maximizing SIC

$$\ell_j - \frac{1}{2} p_j \log n$$

is equivalent to minimizing

$$\log \hat{\sigma}_j^2 + \frac{p_j}{n} \log n$$

It is possible to show that the test statistics,

$$T_n = 2(\ell_j(\hat{\theta}_j) - \ell_i(\hat{\theta}_i)) \rightsquigarrow \chi^2_{p_j - p_i}$$

for $p_j > p_i = p*$. (Note that model i is the true model). Classical hypothesis testing would suggest to reject the null that the smaller model i is the true model if and only if the value of T_n exceeds a $\chi^2_{p_j-p_i}$ critical value. On the other hand, SIC may choose j over i if

$$2\frac{(\ell_j - \ell_i)}{p_j - p_i} > \log n, \tag{1}$$

because

$$\begin{aligned} 0 &= (\ell_j(\hat{\theta}) - \frac{p_j}{2} \log n) - (\ell_i(\hat{\theta}) - \frac{p_i}{2} \log n) \\ &= (\ell_j(\hat{\theta}) - \ell_i(\hat{\theta})) - \frac{1}{2}(p_j - p_i) \log n. \end{aligned}$$

Note that the number of observations is acting like the critical value of the test. This is closely connected with the classical F test, since it is possible to show that, considering for simplicity $p_j - p_i = 1$,

$$\begin{aligned}(\ell_i - \ell_j) &= -n(\log \hat{\sigma}_i^2 - \log \hat{\sigma}_j^2) \\ &= n(\log \hat{\sigma}_j^2 - \log \hat{\sigma}_i^2) \\ &= n \log(\hat{\sigma}_j^2 / \hat{\sigma}_i^2) \\ &= n \log(1 + \hat{\sigma}_j^2 / \hat{\sigma}_i^2 - 1) \\ &= n \log(1 + a)\end{aligned}$$

where $a = (\hat{\sigma}_j^2 / \hat{\sigma}_i^2 - 1)$. Using $\log(1 + a) \approx a$ for small a, we have

$$2(\ell_i - \ell_j) \approx \frac{n(\hat{\sigma}_j^2 - \hat{\sigma}_i^2)}{\hat{\sigma}_i^2} > \log n \tag{2}$$

The LHS of equation (2) can be interpreted as an F test since it can be written approximately as

$$\frac{\mathrm{SSR}_r - \mathrm{SSR}_u}{\mathrm{SSR}_u}.$$

We will use AIC and BIC as model selection devices for time series models. You can find a preliminary example in the tutorial to this lecture notes.

6 A (final) digression

The previous framework connects nicely with Big Data analytic. Consider the following estimator,

$$\hat{\beta} = \arg\min \left[\sum_{i=1}^{n} (y_i - x_i'\beta)^2 + P(\beta, \lambda) \right] \tag{3}$$

where $P(\beta, \lambda)$ is a penalty function and λ is a tuning parameter. For example, the penalty function can be

$$P(\beta, \lambda) = \lambda \sum_{j=1}^{p} |\beta|^q \tag{4}$$

for $q = \{1, 2\}$. The model here is sparse in the coefficients β's and the solution of the optimization problem depends on the value of the shrinkage parameter and the penalty function.

For easily differentiable penalty functions, for instance we have that

$$\begin{aligned}
u'u + \lambda\beta'\beta &= (y - X\beta)'(y - X\beta) + \lambda\beta'\beta \\
&= (y' - \beta'X')(y - X\beta) + \lambda\beta'\beta \\
&= y'y - 2\beta'X' + \beta'X'X\beta + \lambda\beta'\beta
\end{aligned}$$

Assuming $\lambda/2$ given, the FOC $-X'y + X'X\beta + \lambda\beta = 0$. Therefore,

$$\hat{\beta} = (X'X + \lambda I)^{-1}X'y.$$

It can be shown that,

$$\begin{aligned}
E(\hat{\beta}) &= (X'X + \lambda I)^{-1}\beta \\
V(\hat{\beta}) &= (X'X + \lambda I)^{-1}\sigma^2(X'X)(X'X + \lambda I)^{-1}.
\end{aligned}$$

A Appendix:

Recall that $1 = \int f(x)\mathrm{d}x$. Differentiating both sides,

$$\begin{aligned}
0 &= \int \frac{1}{f} f_\theta(x) f \mathrm{d}x \\
0 &= \int \frac{\partial \log f}{\partial \theta} f \mathrm{d}x
\end{aligned}$$

We know that

$$\begin{aligned}
0 &= \int \frac{\partial \log f}{\partial \theta} f \mathrm{d}x \\
0 &= \int \left[\frac{\partial^2 \log f}{\partial \theta \partial \theta'} f + \frac{\partial \log f}{\partial \theta} \frac{\partial f}{\partial \theta} \right] \mathrm{d}x \\
0 &= \int \left[\frac{\partial^2 \log f}{\partial \theta \partial \theta'} f + \left(\frac{\partial \log f}{\partial \theta} \right) \left(\frac{\partial \log f}{\partial \theta'} \right) f \right] \mathrm{d}x \\
0 &= \int \left[\frac{\partial^2 \log f}{\partial \theta \partial \theta'} \right] f dx + \int \left[\left(\frac{\partial \log f}{\partial \theta} \right) \left(\frac{\partial \log f}{\partial \theta'} \right) \right] f \mathrm{d}x
\end{aligned}$$